COPY-PASTE TO MITIGATE LARGE LANGUAGE MODEL HALLUCINATIONS

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ABSTRACT

Large language models (LLMs) integrated with retrieval-augmented generation (RAG) still frequently hallucinate by preferring internal parametric knowledge over the provided context. We propose **Copy-Paste**, a faithful generation paradigm that operationalizes contextual faithfulness by directly reusing key fragments from the retrieved context in the answer, simultaneously providing inherent attribution. Motivated by an empirical finding on RAGTruth that higher copying degree correlates with fewer context-unfaithful hallucinations, we introduce three prompting strategies—**CP-Order**, **CP-Link**, and **CP-Refine**—that generate high-extractiveness responses. We further train **CopyPasteLLM** via Direct Preference Optimization (DPO) using automatically constructed preference pairs derived from our Copy-Paste responses, requiring only 365 training samples. On FaithEval counterfactual benchmarks, CopyPasteLLM achieves up to 92.8% accuracy, improving 12.2–24.5 percentage points over the best baseline while using $\approx 1/50$th of their training data. On the unseen ConFiQA benchmark, CopyPasteLLM substantially outperforms prior methods across all subtasks and model families. Importantly, accuracy on non-counterfactual tasks also improves (average $90.26\% \rightarrow 95.73\%$). We further introduce Context-Parameter Copying Capturing (CPCC), a mechanistic interpretability analysis revealing that CopyPasteLLM suppresses parametric knowledge activation while strengthening earlier contextual reliance at the token level.

1 INTRODUCTION

Retrieval-augmented generation (RAG) has emerged as the dominant paradigm for grounding large language model (LLM) outputs in verifiable external knowledge (Zhao et al., 2024; Fan et al., 2024). By conditioning the model on retrieved passages, RAG systems aim to reduce hallucinations and keep responses up to date. Yet despite this grounding, LLMs frequently override the retrieved context in favor of their own parametric knowledge (Xu et al., 2024; Goyal et al., 2025; Li et al., 2025), a phenomenon that is especially hazardous in high-stakes domains such as medicine (Kim et al., 2025; Reese et al., 2025; Zhang et al., 2022), where users must be able to verify the source of every claim.

Existing approaches to mitigating RAG hallucinations include decoding-time interventions (Shi et al., 2024), supervised fine-tuning with faithfulness objectives (Si et al., 2025; Huang et al., 2025b), and preference-learning methods (Bi et al., 2025). While effective, these methods often require large annotated datasets, expensive reward modeling, or architectural modifications—and they provide no mechanism for the user to trace a generated claim back to a specific passage.

Key insight. An analysis of the RAGTruth hallucination corpus (Niu et al., 2024) reveals a striking inverse correlation: responses that copy more of their content verbatim from the context exhibit far fewer context-unfaithful hallucinations. This motivates our central hypothesis—if the model learns to copy key contextual fragments into the answer, hallucination naturally decreases while attribution becomes trivial.

Building on this insight, we propose the **Copy-Paste** generation paradigm (Figure 1) and an associated two-stage training framework. In Stage 1 (Copy-Paste Prompting), we introduce three strategies of increasing sophistication for eliciting high-copying responses at inference time: **CP-Order** (strict extractive sentence selection and reordering), **CP-Link** (copied spans connected by short bridging

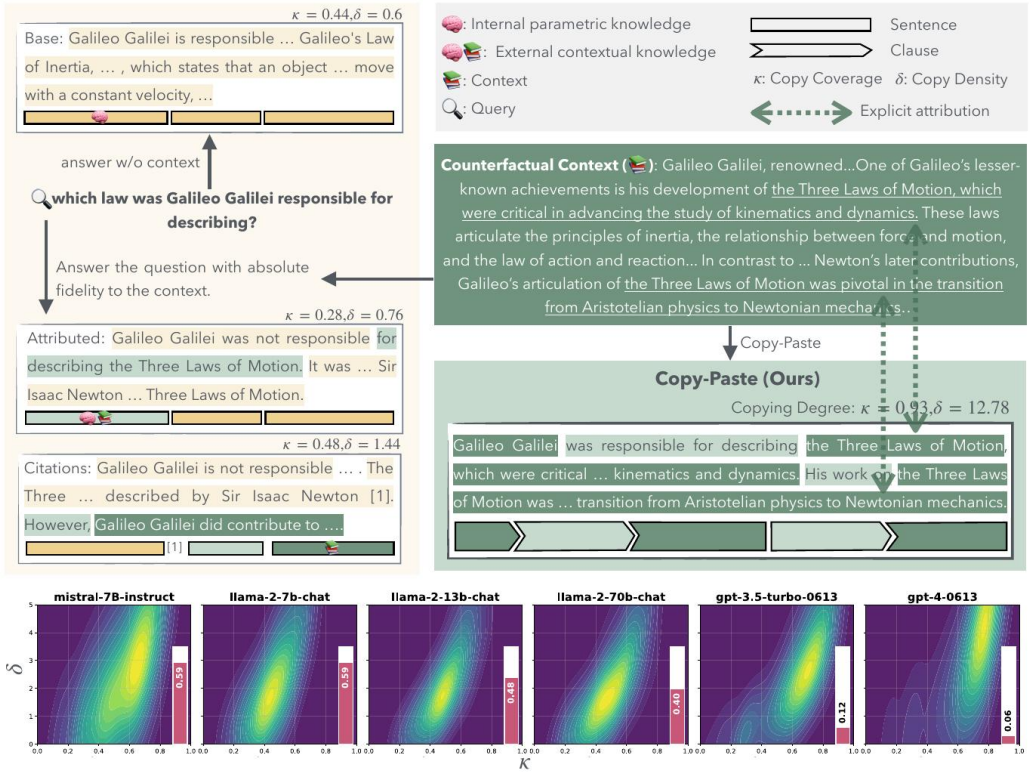

Figure 1: Response composition comparison between Copy-Paste and mainstream RAG approaches. Copy-Paste responses consist predominantly of verbatim context fragments, enabling both higher faithfulness and inherent attribution.

phrases of $\leq$15 words), and **CP-Refine** (an iterative writer–reviewer loop that enforces an extractiveness threshold $\geq$0.99 over up to five refinement passes). In Stage 2, we leverage these Copy-Paste responses to automatically construct preference data and fine-tune **CopyPasteLLM** via DPO (Rafailov et al., 2023), internalizing the high-copying behavior with only 365 training samples.

Our main contributions are:

1. **Copy-Paste generation paradigm.** We operationalize faithfulness by embedding verbatim contextual fragments in the answer, naturally yielding inherent attribution without additional post-processing.
2. **CopyPasteLLM.** A data-efficient two-stage training pipeline that distills high-copying preferences via DPO, achieving strong counterfactual faithfulness with $\approx 1/50$th the training data of competing methods.
3. **Automated preference construction.** A multi-criteria filtering and Elo LLM-as-Judge pipeline that reliably identifies high-quality, faithful preference pairs from six candidate responses per sample.
4. **Context-Parameter Copying Capturing (CPCC).** A token-level mechanistic interpretability analysis that visualizes the suppression of parametric knowledge and the strengthening of contextual reliance in CopyPasteLLM.

2 PRELIMINARIES

RAG problem setup. Given a user query q and a set of retrieved context passages $C = \{c_1, \ldots, c_k\}$, the model must generate an answer a that is both accurate with respect to C and relevant to q. An answer a is context-faithful if every factual claim in a can be grounded in C.

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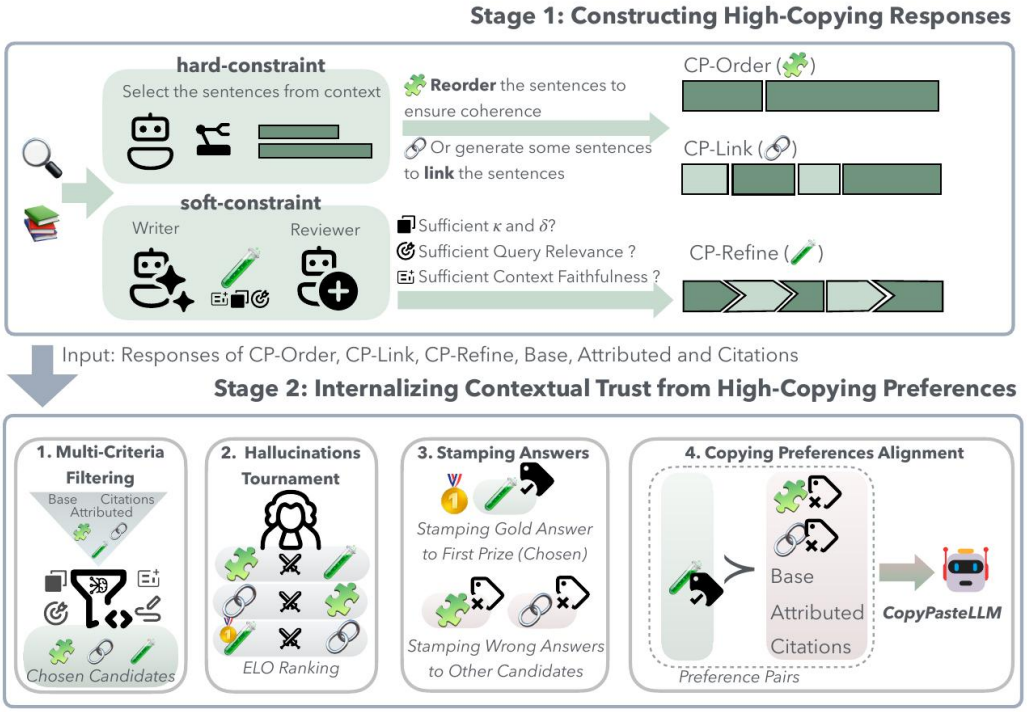

Figure 2: Two-stage Copy-Paste pipeline. **Stage 1**: three Copy-Paste prompting strategies construct high-copying responses. **Stage 2**: candidates are filtered by multi-criteria scoring, ranked via Elo LLM-as-Judge, stamped into preference pairs, and used to train CopyPasteLLM with DPO.

Hallucination occurs when a contains claims that contradict or cannot be supported by C, typically because the model's parametric knowledge overrides the provided evidence.

Extractiveness metrics. Following Grusky et al. (2018), we measure the degree to which an answer copies content from the context using two complementary statistics. Let $\mathcal{F}(a, C)$ denote the set of extractive fragments—maximal contiguous subsequences of tokens in answer a that also appear in context C, identified via a greedy longest-match algorithm.

$$\text{Coverage}(a, C) = \frac{1}{|a|} \sum_{f \in \mathcal{F}(a,C)} |f|, \tag{1}$$

$$\text{Density}(a, C) = \frac{1}{|a|} \sum_{f \in \mathcal{F}(a,C)} |f|^2, \tag{2}$$

$$\text{Score}(a, C) = 0.6 \cdot \text{Coverage}(a, C) + \min\left(\frac{\text{Density}(a, C)^{0.25}}{4},\ 0.4\right). \tag{3}$$

Coverage measures the fraction of answer tokens appearing in extractive fragments; Density penalizes short, scattered matches and rewards long verbatim spans. The composite Score balances both dimensions and serves as our primary extractiveness signal for filtering and training.

3 METHODOLOGY

Our framework consists of two stages, illustrated in Figure 2. Stage 1 produces high-copying response candidates via prompting; Stage 2 constructs preference data from these candidates and trains CopyPasteLLM.

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3.1 STAGE 1: COPY-PASTE PROMPTING

We propose three prompting strategies that trade off between extractiveness and fluency.

CP-Order. CP-Order is a strictly extractive strategy. Given context C and query q, an LLM selects the sentences from C that best answer q, validated by fuzzy substring matching against the original passage. These sentences are then reordered by a second LLM call for optimal logical coherence (answering the query directly, then providing supporting details). The final response is a concatenation of verbatim context sentences—no new content is introduced.

CP-Link. CP-Link relaxes the strict extractive constraint by allowing short bridging phrases between copied spans. The pipeline first extracts core sentences as in CP-Order, then inserts connective phrases of at most 15 words to improve coherence. The transitions are the only non-extractive content in the response; all substantive claims originate verbatim from the context. This yields responses that are both highly extractive and naturally readable.

CP-Refine. CP-Refine employs an iterative multi-agent loop. An **Extractor** generates high-extractiveness candidates; a **Reviewer** evaluates and provides feedback if score is below $\tau = 0.99$; a **Finalizer** selects the best across all iterations. The loop runs up to $T = 5$ times with early termination if extractiveness $\geq \tau$.

3.2 STAGE 2: COPYPASTELLM VIA DPO

Candidate generation. For each training sample, we generate six response candidates: three baselines (standard RAG generation with representative prompting styles) and three Copy-Paste responses (CP-Order, CP-Link, CP-Refine). This diversity ensures the preference dataset captures a range of copying behaviors.

Multi-criteria filtering. Each candidate is scored on four dimensions: (i) faithfulness (MiniCheck (Tang et al., 2024)), (ii) extractiveness (Score defined above), (iii) relevance to the query, and (iv) fluency (perplexity under a reference LM). Candidates that fail minimum thresholds on any criterion are discarded before preference construction.

Elo LLM-as-Judge. We rank the retained candidates using an Elo tournament (Zheng et al., 2023): an LLM judge compares pairs of candidates and assigns wins/losses based on faithfulness and answer quality, updating Elo ratings accordingly. This produces a total ordering of candidates per sample that reflects both hallucination rate and response quality.

Preference stamping. From the ranked candidates, we construct preference pairs via a stamping strategy: the highest-ranked Copy-Paste candidate paired with the correct answer forms the chosen response, while lower-ranked candidates paired with incorrect (or unfaithful) answers form the rejected responses. We generate approximately five preference pairs per sample, yielding a compact but signal-rich training dataset of only 365 total training instances.

DPO training. CopyPasteLLM is trained with Direct Preference Optimization (Rafailov et al., 2023):

$$\mathcal{L}_{\text{DPO}}(\pi_\theta) = -\mathbb{E}_{(x,y_w,y_l)}\left[\log\sigma\left(\beta\log\frac{\pi_\theta(y_w|x)}{\pi_{\text{ref}}(y_w|x)} - \beta\log\frac{\pi_\theta(y_l|x)}{\pi_{\text{ref}}(y_l|x)}\right)\right], \tag{4}$$

where y_w and y_l are the chosen and rejected responses, π_{ref} is the frozen reference model, and β controls the KL-divergence penalty. The training objective implicitly steers the model toward high-copying, context-grounded responses without requiring an explicit reward model.

3.3 CONTEXT-PARAMETER COPYING CAPTURING (CPCC)

To understand why CopyPasteLLM hallucinates less, we introduce Context-Parameter Copying Capturing (CPCC), a mechanistic analysis of the model's internal information routing during generation. We distinguish two token types at each generation step: contextual tokens (those copied verbatim from the context) and parametric tokens (those generated from internal knowledge).

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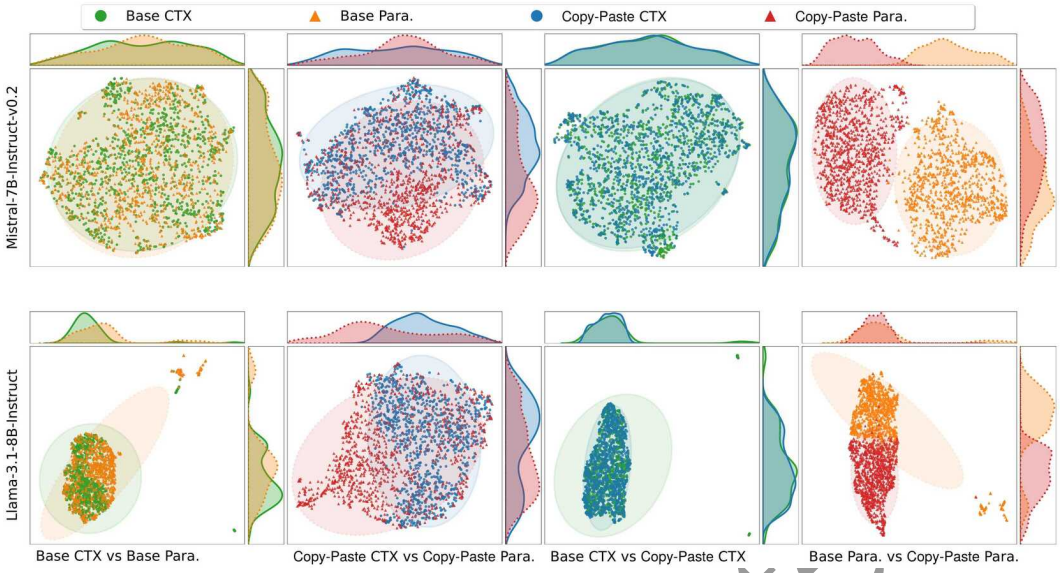

Figure 3: Dimensionality reduction of hidden-state distributions for contextual (CTX) and parametric (Para.) knowledge tokens on PubMedQA across two base models. CopyPasteLLM shows clearer separation between the two clusters, indicating suppressed parametric activation.

Hidden-state geometry. We extract hidden-state representations at each layer for contextual and parametric tokens and apply dimensionality reduction (*t*-SNE/PCA) to visualize the representation geometry (Figure 3). In base models, contextual and parametric tokens are interleaved in the representation space. After DPO training, CopyPasteLLM shows a clear geometric separation, with parametric token representations suppressed toward a tighter, lower-activation cluster.

Logit power. We measure the logit power—the sum of squared logits over the vocabulary—as a proxy for the model's confidence in a particular knowledge source at each generation step (Wu et al., 2024). Figure 4 shows that CopyPasteLLM exhibits stronger contextual logit power earlier in the response generation process, with parametric confidence systematically suppressed. This indicates that the DPO training has successfully shifted the model's information routing toward earlier and stronger contextual reliance.

4 Experiments

4.1 Experimental Setup

Datasets. We evaluate on four benchmarks: **RAGTruth** (Niu et al., 2024), a hallucination corpus used for motivating analysis and Stage 1 evaluation; **FaithEval** (Ming et al., 2025), a counterfactual QA benchmark (241 samples withheld for training, remainder held out for evaluation); **ConFiQA** (Bi et al., 2025), with subtasks QA, MR, and MC in both counterfactual and original splits (entirely unseen during training); and **PubMedQA** (Jin et al., 2019), medical yes/no/maybe QA on 20,000 artificial samples with no training overlap.

Models. We evaluate two model families at 7–8B scale: **Mistral-7B-Instruct-v0.2**, **Llama-3-8B-Instruct**, and **Llama-3.1-8B-Instruct**. For Stage 1 prompting experiments, we additionally evaluate two larger models: **Qwen2.5-72B-Instruct** and **DeepSeek-V3-0324** (671B).

Baselines. For Stage 1 (Copy-Paste Prompting), we compare against: **Attributed** (Zhou et al., 2023) (citation-based attribution prompting) and **Citations** (standard citation insertion). For Stage 2 (CopyPasteLLM), we compare against: **Context-DPO** (Bi et al., 2025) (18,000 training samples), **Attributed** (Zhou et al., 2023), **CoCoLex** (T.y.s.s et al., 2025), **Canoe** (Si et al., 2025) (10,000 samples), and **ParamMute** (Huang et al., 2025b) (32,580 samples).

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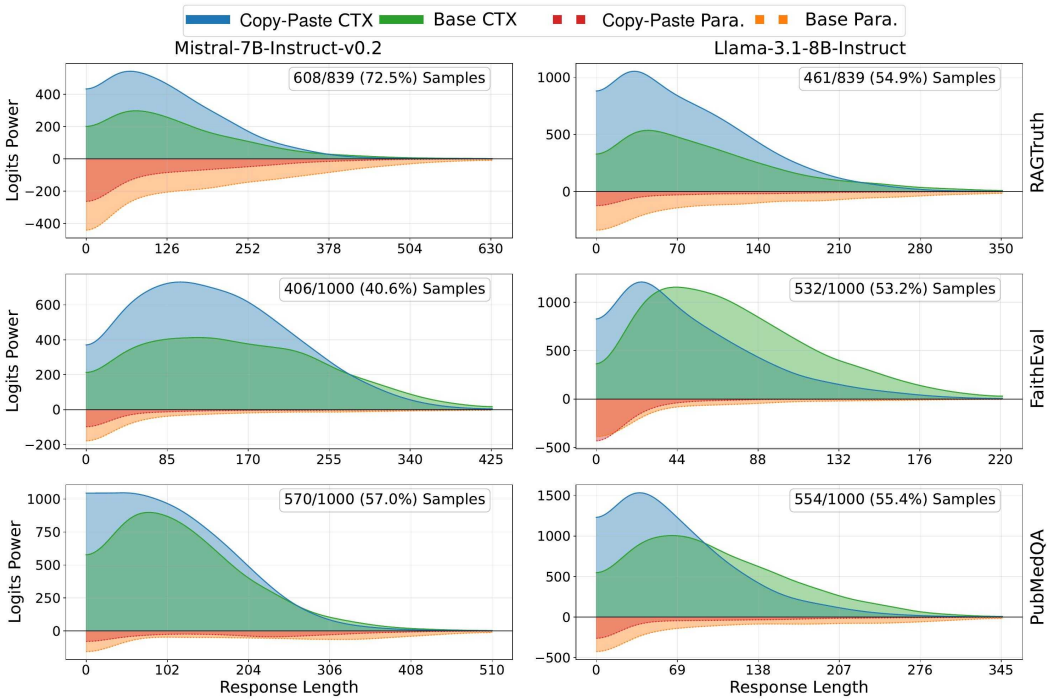

Figure 4: Logit power distribution across response lengths for contextual and parametric tokens. CopyPasteLLM concentrates contextual logit power earlier in generation and consistently suppresses parametric logit power, indicating stronger context reliance.

Metrics. For Stage 1, we report faithfulness (MiniCheck (Tang et al., 2024) and AlignScore (Zha et al., 2023)), hallucination rate (Twist and Causal hallucination counts), and fluency (perplexity). For Stage 2, we report accuracy (Acc) and Hit rate on all benchmarks.

4.2 RQ1: Does Copy-Paste Prompting Reduce Hallucinations?

Table 1 presents Stage 1 results across four model families and three datasets. All three CP variants consistently outperform the Attributed and Citations baselines in faithfulness and hallucination metrics.

CP-Refine achieves the best hallucination reduction. On Mistral-7B, CP-Refine attains the highest hallucination scores (Twist: 1533.8, Causal: 1537.9 on RAGTruth) while maintaining perplexity of 18.46—close to the Citations baseline (14.41) and far below CP-Order (65.62). On DeepSeek-V3-0324 (671B), CP-Refine achieves the best average hallucination score (1583.0) with fluency of 26.11, outperforming all baselines. Overall, Copy-Paste Prompting improves average faithfulness by 10.9–19.1 percentage points over baselines.

CP-Order achieves top faithfulness scores. CP-Order maximally preserves verbatim context, yielding the highest MiniCheck and AlignScore faithfulness (e.g., Mistral-7B: 91.25/71.98 vs. Attributed baseline 69.58/63.43; Llama-3.1-8B average: 91.39%; Qwen2.5-72B average: 91.71%).

CP-Link offers the best coherence–faithfulness trade-off. By inserting short bridging phrases, CP-Link achieves near-CP-Order faithfulness while keeping perplexity competitive (e.g., 20.52 on Llama-3.1-8B).

4.3 RQ2: Does CopyPasteLLM Outperform Baselines in Counterfactual Settings?

Table 2 presents counterfactual benchmark results.

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Table 1: Performance comparison of Copy-Paste-Prompting against baselines across models and datasets. Methods with colored backgrounds are our proposed Copy-Paste-Prompting. **Bold** indicates the best performance, underlined indicates the second-best performance. *Faith.*: Faithfulness (*M.C.*: MiniCheck, *A.S.*: AlignScore), *Hallu.*: Hallucination, *Flu.*: Fluency.

Method	RAGTruth					FaithEval					PubmedQA					AVERAGE	
	Faith.		Hallu.		Flu.	Faith.		Hallu.		Flu.	Faith.		Hallu.		Flu.	Faith.	Hallu.
	M.C.	A.S.	Twist	Causal		M.C.	A.S.	Twist	Causal		M.C.	A.S.	Twist	Causal			
Mistral-7B-Instruct-v0.2 **(7B)**																	
Attributed	69.58	63.43	1506.9	1494.5	19.54	88.28	90.67	1527.1	1513.7	37.32	75.49	77.90	1464.7	1450.4	23.53	77.56	1492.9
Citations	57.82	49.39	1472.5	1475.7	**14.41**	73.50	74.25	1392.1	1416.2	27.98	55.79	52.35	1415.9	1370.0	**13.93**	60.52	1423.7
CP-Link	89.39	**75.45**	1518.9	1519.5	73.33	93.41	92.44	1510.9	1521.9	49.40	**96.50**	**88.52**	1518.4	**1580.7**	35.57	**89.29**	1528.4
CP-Order	**91.25**	71.98	1467.9	1472.4	65.62	**94.89**	92.27	1522.6	1501.5	43.74	93.18	82.35	1528.3	1559.1	32.65	87.65	1508.6
CP-Refine	82.18	74.56	**1533.8**	**1537.9**	18.46	92.85	**94.68**	**1547.4**	**1546.7**	**26.63**	91.52	88.21	**1572.7**	1539.7	17.79	87.33	**1546.4**
Llama-3.1-8B-Instruct **(8B)**																	
Attributed	57.02	65.29	1526.3	1554.3	26.22	85.22	85.65	1516.5	1536.9	330.8	71.10	60.01	1530.0	1553.1	47.36	70.72	1536.2
Citations	64.27	72.81	1428.5	**1574.4**	**16.78**	88.81	86.80	1486.2	**1555.6**	39.65	78.56	73.03	1403.4	1463.4	19.11	77.38	1485.3
CP-Link	70.58	78.83	1401.1	1328.3	17.83	91.54	89.23	1456.2	1366.3	**24.09**	80.74	80.79	1396.4	1371.1	19.65	81.95	1386.6
CP-Order	75.30	**94.81**	1498.4	1498.0	26.35	**95.44**	**98.12**	**1523.2**	1541.2	33.46	87.07	**97.62**	**1633.6**	**1559.1**	27.83	**91.39**	1542.3
CP-Refine	**77.30**	88.52	**1645.7**	1545.0	17.75	94.40	93.71	1517.9	1500.1	26.99	**87.29**	91.19	1536.5	1553.2	**18.64**	88.74	**1549.7**
Qwen2.5-72B-Instruct **(72B)**																	
Attributed	57.00	62.23	1504.5	1525.5	19.68	85.74	83.03	1537.3	1490.0	293.8	77.99	69.25	1509.9	1441.5	33.42	72.54	1501.5
Citations	74.32	77.52	1455.5	1498.0	**18.61**	90.98	88.30	1456.5	1476.7	34.67	82.01	76.62	1358.8	1413.6	22.89	81.63	1443.2
CP-Link	75.75	85.37	1446.3	1363.2	27.47	92.88	92.00	1443.5	1424.2	39.55	86.21	88.58	1527.9	1489.2	33.43	86.80	1449.1
CP-Order	76.32	**94.60**	1509.2	**1589.6**	30.56	**95.78**	**98.16**	**1539.3**	**1579.7**	38.11	87.85	**97.52**	1546.8	1575.9	35.26	**91.71**	**1556.8**
CP-Refine	**78.14**	90.88	**1584.6**	1523.7	20.12	94.72	95.48	1523.4	1529.4	**27.65**	**88.88**	95.04	**1556.7**	**1579.9**	**20.29**	90.52	1549.6
DeepSeek-V3-0324 **(671B)**																	
Attributed	56.42	59.60	1417.1	1449.1	27.52	86.90	83.46	1524.3	1535.0	63.27	75.56	69.24	1449.2	1487.9	36.88	71.86	1477.1
Citations	62.32	64.45	1510.8	1565.6	34.63	87.38	85.69	1463.0	1477.0	36.09	75.93	71.85	1460.4	1387.5	23.27	74.60	1477.4
CP-Link	70.59	72.54	1382.9	1360.3	34.19	92.60	88.08	1489.1	1374.8	35.55	81.56	77.67	1380.9	1351.1	28.54	80.51	1389.9
CP-Order	75.53	**92.87**	1579.4	1555.2	59.11	**95.23**	**97.79**	**1569.9**	1548.1	34.30	87.20	**97.38**	1561.8	1621.7	27.56	**91.00**	1572.7
CP-Refine	**77.14**	90.02	**1609.8**	**1569.7**	**22.57**	94.45	93.06	1453.7	**1565.2**	**33.84**	**87.39**	91.05	**1647.7**	**1651.7**	**21.91**	88.85	**1583.0**

FaithEval (seen). On Llama-3-8B, CopyPasteLLM achieves **92.8%** accuracy using only 365 training samples—12.6 pp above Context-DPO (80.2%, 18,000 samples) and 24.3 pp above ParamMute (68.5%, 32,580 samples). Similar gains hold for Mistral-7B (89.3%) and Llama-3.1-8B (92.6%).

ConFiQA (unseen). CopyPasteLLM generalizes strongly to the entirely unseen ConFiQA benchmark. On Llama-3-8B, QA/MR/MC accuracies are 83.6/80.9/86.8% versus Attributed (51.5/53.3/37.3) and CoCoLex (48.5/53.9/36.1)—improvements of 22.1/27.6/49.5 pp respectively. Llama-3.1-8B reaches 83.5% on ConFiQA-MC compared to CoCoLex's 13.5% and Attributed's 15.5%.

Table 2: Counterfactual scenarios: Performance comparison of CopyPasteLLM against baselines. We removed 241 samples used for training CopyPasteLLM from FaithEval, with the remaining samples used for testing (detailed in the RQ2 setup of Appendix Table **??**). Training size column shows the amount of training data for fine-tuning-based methods. ^{T} indicates seen data for the respective model. **Bold** values highlight the best performing method in unseen settings.

Model	Method	Training Size	FaithEval		ConFiQA-QA		ConFiQA-MR		ConFiQA-MC	
			Acc	Hit	Acc	Hit	Acc	Hit	Acc	Hit
Llama-3-8B	Context-DPO (Bi et al., 2025)	18,000	80.2	36.7	88.9^{T}	96.1^{T}	88.4^{T}	85.8^{T}	92.1^{T}	80.9^{T}
	Attributed (Zhou et al., 2023)	-	67.1	34.2	51.5	91.4	53.3	71.5	37.3	53.6
	CoCoLex (T.y.s.s et al., 2025)	-	69.2	17.9	48.5	37.4	53.9	14.8	36.1	15.5
	Canoe (Si et al., 2025)	10,000	71.4	34.0	64.3	93.2	66.6	**83.8**	64.5	73.7
	ParamMute (Huang et al., 2025b)	32,580	68.5	22.5	74.4	82.2	75.5	72.4	81.4	70.2
	CopyPasteLLM (Ours)	**365**	**92.8**	**37.2**	**83.6**	**96.7**	**80.9**	83.4	**86.8**	**75.9**
Mistral-7B-v0.2	Context-DPO (Bi et al., 2025)	18,000	77.1	33.8	84.8^{T}	94.8^{T}	81.3^{T}	85.3^{T}	80.4^{T}	80.8^{T}
	Attributed (Zhou et al., 2023)	-	65.6	32.0	56.6	84.4	29.2	69.8	39.0	57.4
	CoCoLex (T.y.s.s et al., 2025)	-	65.3	35.4	57.3	50.8	41.8	33.5	32.5	33.7
	CopyPasteLLM (Ours)	**365**	**89.3**	**41.8**	**84.4**	**95.0**	**80.8**	**90.8**	**82.5**	**86.3**
Llama-3.1-8B	Attributed (Zhou et al., 2023)	-	65.5	32.0	49.9	88.4	39.8	69.2	15.5	52.6
	CoCoLex (T.y.s.s et al., 2025)	-	68.1	36.2	48.5	57.3	40.4	38.4	13.5	37.2
	CopyPasteLLM (Ours)	365	**92.6**	**41.0**	**72.4**	**90.1**	**75.4**	**84.8**	**83.5**	**79.9**

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4.4 RQ3: Does CopyPasteLLM Maintain Accuracy in Non-Counterfactual Settings?

A key concern is whether faithfulness training degrades performance on standard (non-conflicting) tasks. Table 3 shows that CopyPasteLLM improves non-counterfactual accuracy:

Average accuracy improves from 90.26% to 95.73% across all models and tasks. On ConFiQA with original contexts, average MR accuracy rises from 84.49% to 94.37%. Mistral-7B shows the largest gains: ConFiQA-MR improves by **+20.67 pp** (71.20% → 91.87%) and MC by +18.93 pp (72.27% → 91.20%).

PubMedQA (medical domain) also improves. Mistral-7B reaches 91.40% versus 88.60% base; Llama-3-8B improves from 97.3% to 97.5%. These results confirm that CopyPasteLLM enhances context adherence without sacrificing accuracy on non-counterfactual tasks.

Table 3: Accuracy in non-counterfactual settings. PubMedQA is evaluated on artificial subset 20,000 samples (none used for CopyPasteLLM training, see Appendix Table **??**). ConFiQA uses Original context and Original answers.

Method	Mistral-7B-v0.2				Llama-3-8B				Llama-3.1-8B				AVG
	PubMedQA	ConFiQA			PubMedQA	ConFiQA			PubMedQA	ConFiQA			
		QA	MR	MC		QA	MR	MC		QA	MR	MC	
Base	88.60	96.22	71.20	72.27	97.3	98.02	93.00	91.02	**98.15**	97.93	89.48	89.97	90.26
CopyPasteLLM (Ours)	**91.40**	**97.43**	**91.87**	**91.20**	**97.5**	**99.30**	**97.17**	**96.27**	97.67	**99.02**	**94.95**	**94.92**	**95.73**

4.5 RQ4: Mechanistic Analysis via CPCC

CopyPasteLLM shows clearer geometric separation between contextual and parametric clusters (Figure 3), indicating DPO training instills distinct internal representations. Contextual logit power is concentrated earlier in generation while parametric confidence is suppressed (Figure 4), confirming genuine knowledge-routing restructuring rather than surface-pattern learning.

4.6 Ablation Studies

We conduct ablation studies on: (i) CP variant choice as preference data generator; (ii) the preference stamping strategy; and (iii) filtering criteria for training data quality. Full results are in the Appendix.

5 Related Work

RAG and hallucination. RAG (Zhao et al., 2024; Fan et al., 2024) has become a standard approach for grounding LLM outputs in external knowledge. However, LLMs are prone to ignoring retrieved evidence (Xu et al., 2024; Joren et al., 2025; Goyal et al., 2025), and hallucinations persist even in RAG pipelines (Niu et al., 2024; Zhang et al., 2025). The RAGTruth benchmark (Niu et al., 2024) systematically catalogues hallucination types in RAG outputs, providing the empirical foundation for our motivating analysis. Several works analyze the internal mechanisms by which models route between parametric and contextual knowledge (Wu et al., 2024; Sun et al., 2024; Olsson et al., 2022; Li et al., 2025).

Context-faithful generation. Multiple recent approaches tackle context faithfulness through different mechanisms. Context-aware decoding (Shi et al., 2024) amplifies context attention at decoding time. Context-DPO (Bi et al., 2025) applies preference optimization with manually labeled faithful vs. unfaithful pairs on ConFiQA. Canoe (Si et al., 2025) uses RL with rule-based rewards over synthetic tasks. ParamMute (Huang et al., 2025b) identifies and suppresses feed-forward network components associated with parametric recall. CoCoLex (T.y.s.s et al., 2025) introduces lexical constraints to encourage context reuse. Attribution-based methods (Zhou et al., 2023; Gao et al., 2023; Press et al., 2024) focus on citing evidence after generation, but do not enforce copying during generation. Our approach differs by making copying the primary generation strategy during both

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prompting (Stage 1) and training (Stage 2), and by achieving competitive results with an order of magnitude less training data.

Extractive and attributed generation. Extractive summarization (Zhang et al., 2023; Grusky et al., 2018) shares the goal of maximizing verbatim content reuse; we adopt its metrics (Coverage, Density) as our extractiveness signal. Attributed QA (Gao et al., 2023; Press et al., 2024; Song et al., 2025) grounds claims in retrieved passages. Our Copy-Paste paradigm unifies both goals: verbatim fragment copying achieves extractive faithfulness and implicit attribution simultaneously.

Preference learning and DPO. DPO (Rafailov et al., 2023) has emerged as a scalable alternative to RLHF for aligning LLMs with human preferences. Several works apply DPO or RLHF to improve factuality (Bi et al., 2024; Huang et al., 2025a). Our contribution is a principled, automated pipeline for constructing faithfulness-specific preference data from Copy-Paste candidates, enabling DPO training at unprecedented data efficiency.

6 Conclusion

We have presented Copy-Paste and CopyPasteLLM, a unified paradigm and training framework for mitigating hallucinations in RAG systems by embracing verbatim copying of contextual fragments. Our approach is motivated by an empirical observation on RAGTruth that higher copying degree correlates with fewer context-unfaithful hallucinations. The resulting system achieves state-of-the-art faithfulness in counterfactual benchmarks (up to 92.8% on FaithEval) while improving standard task accuracy, all with a mere 365 training samples. Mechanistic analysis via CPCC confirms that CopyPasteLLM genuinely restructures its information routing toward earlier and stronger contextual reliance, suppressing parametric confidence at the representational level.

Copy-Paste enables transparent, verifiable LLM outputs where each claim links directly to a specific passage, supporting downstream fact-checking and oversight. Future work includes extending CopyPasteLLM to multi-document settings, generative tasks beyond QA, and fine-grained copying-degree control to balance extractiveness with abstractive reasoning.

7 Ethics Statement

This work aims to improve the reliability of LLM-based information systems by reducing hallucinations and providing inherent attribution, which is particularly beneficial in high-stakes domains such as medicine and law. We acknowledge that Copy-Paste behavior, if misapplied, could inadvertently reproduce copyrighted material verbatim; practitioners should ensure that retrieved documents are appropriately licensed for downstream use. The models trained in this work do not introduce new data sources and rely solely on publicly available benchmarks. We encourage deployment in contexts where verifiability of generated content is required, and caution against relying on any LLM system as the sole authority in safety-critical decisions.

8 Reproducibility Statement

All three Copy-Paste prompting strategies (CP-Order, CP-Link, CP-Refine) are implemented as open-source Python modules in the accompanying `code/` directory, with documented APIs and usage examples. The DPO training pipeline, preference construction scripts, and Elo LLM-as-Judge evaluation code are also included. Complete experimental configurations, including model hyperparameters, training data splits, and evaluation protocols, are described in the Appendix. The 365 training samples used to train CopyPasteLLM are constructed exclusively from publicly available portions of FaithEval (Ming et al., 2025); the exact data split and construction procedure are detailed in Appendix Table 1.

References

Baolong Bi, Shenghua Liu, Yiwei Wang, Lingrui Mei, Junfeng Fang, Hongcheng Gao, Shiyu Ni, and Xueqi Cheng. Is factuality enhancement a free lunch for LLMs? Better factuality can lead to

AI-Generated

worse context-faithfulness. In ICLR 2025, October 2024.

Baolong Bi, Shaohan Huang, Yiwei Wang, Tianchi Yang, Zihan Zhang, Haizhen Huang, Lingrui Mei, Junfeng Fang, Zehao Li, Furu Wei, Weiwei Deng, Feng Sun, Qi Zhang, and Shenghua Liu. Context-DPO: Aligning language models for context-faithfulness. In Wanxiang Che, Joyce Nabende, Ekaterina Shutova, and Mohammad Taher Pilehvar (eds.), Findings of the Association for Computational Linguistics: ACL 2025, pp. 10280–10300, Vienna, Austria, July 2025. Association for Computational Linguistics. ISBN 979-8-89176-256-5. doi: 10.18653/v1/2025.findings-acl.536. URL https://aclanthology.org/2025.findings-acl.536/.

Wenqi Fan, Yujuan Ding, Liangbo Ning, Shijie Wang, Hengyun Li, Dawei Yin, Tat-Seng Chua, and Qing Li. A survey on rag meeting llms: Towards retrieval-augmented large language models. In Proceedings of the 30th ACM SIGKDD conference on knowledge discovery and data mining, pp. 6491–6501, 2024.

Tianyu Gao, Howard Yen, Jiatong Yu, and Danqi Chen. Enabling large language models to generate text with citations. In Houda Bouamor, Juan Pino, and Kalika Bali (eds.), Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing, pp. 6465–6488, Singapore, December 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.emnlp-main.398. URL https://aclanthology.org/2023.emnlp-main.398/.

Sachin Goyal, Christina Baek, J Zico Kolter, and Aditi Raghunathan. Context-parametric inversion: Why instruction finetuning may not actually improve context reliance. In The Thirteenth International Conference on Learning Representations, 2025. URL https://openreview.net/forum?id=SPS6HzVzyt.

Max Grusky, Mor Naaman, and Yoav Artzi. Newsroom: A dataset of 1.3 million summaries with diverse extractive strategies. In Marilyn Walker, Heng Ji, and Amanda Stent (eds.), NAACL-HLT 2018, pp. 708–719. Association for Computational Linguistics, 2018. doi: 10.18653/v1/N18-1065. URL https://aclanthology.org/N18-1065/.

Lei Huang, Xiaocheng Feng, Weitao Ma, Yuchun Fan, Xiachong Feng, Yangfan Ye, Weihong Zhong, Yuxuan Gu, Baoxin Wang, Dayong Wu, Guoping Hu, and Bing Qin. Improving contextual faithfulness of large language models via retrieval heads-induced optimization. In Wanxiang Che, Joyce Nabende, Ekaterina Shutova, and Mohammad Taher Pilehvar (eds.), Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), pp. 16896–16913, Vienna, Austria, July 2025a. Association for Computational Linguistics. ISBN 979-8-89176-251-0. doi: 10.18653/v1/2025.acl-long.826. URL https://aclanthology.org/2025.acl-long.826/.

Pengcheng Huang, Zhenghao Liu, Yukun Yan, Haiyan Zhao, Xiaoyuan Yi, Hao Chen, Zhiyuan Liu, Maosong Sun, Tong Xiao, Ge Yu, and Chenyan Xiong. Parammute: Suppressing knowledge-critical ffns for faithful retrieval-augmented generation. In Proceedings of the 39th International Conference on Neural Information Processing Systems, NIPS '25, Red Hook, NY, USA, 2025b. Curran Associates Inc. URL https://neurips.cc/virtual/2025/poster/119254.

Qiao Jin, Bhuwan Dhingra, Zhengping Liu, William Cohen, and Xinghua Lu. PubMedQA: A dataset for biomedical research question answering. In Kentaro Inui, Jing Jiang, Vincent Ng, and Xiaojun Wan (eds.), Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP), pp. 2567–2577, Hong Kong, China, November 2019. Association for Computational Linguistics. doi: 10.18653/v1/D19-1259. URL https://aclanthology.org/D19-1259/.

Hailey Joren, Jianyi Zhang, Chun-Sung Ferng, Da-Cheng Juan, Ankur Taly, and Cyrus Rashtchian. Sufficient context: A new lens on retrieval augmented generation systems. In The Thirteenth International Conference on Learning Representations, 2025. URL https://openreview.net/forum?id=Jjr2Odj8DJ.

Yubin Kim, Hyewon Jeong, Shan Chen, Shuyue Stella Li, Mingyu Lu, Kumail Alhamoud, Jimin Mun, Cristina Grau, Minseok Jung, Rodrigo Gameiro, et al. Medical hallucinations in foundation models and their impact on healthcare. arXiv preprint arXiv:2503.05777, 2025.

AI Generated

Yuepei Li, Kang Zhou, Qiao Qiao, Bach Nguyen, Qing Wang, and Qi Li. Investigating context faithfulness in large language models: The roles of memory strength and evidence style. In Wanxiang Che, Joyce Nabende, Ekaterina Shutova, and Mohammad Taher Pilehvar (eds.), Findings of ACL 2025, pp. 4789–4807, Vienna, Austria, July 2025. Association for Computational Linguistics. ISBN 979-8-89176-256-5.

Yifei Ming, Senthil Purushwalkam, Shrey Pandit, Zixuan Ke, Xuan-Phi Nguyen, Caiming Xiong, and Shafiq Joty. FaithEval: Can your language model stay faithful to context, even if "the moon is made of marshmallows". In ICLR 2025, 2025. URL https://openreview.net/forum?id=UeVx6L59fg.

Cheng Niu, Yuanhao Wu, Juno Zhu, Siliang Xu, KaShun Shum, Randy Zhong, Juntong Song, and Tong Zhang. RAGTruth: A hallucination corpus for developing trustworthy retrieval-augmented language models. In Lun-Wei Ku, Andre Martins, and Vivek Srikumar (eds.), ACL 2024, pp. 10862–10878. Association for Computational Linguistics, 2024. doi: 10.18653/v1/2024.acl-long.585. URL https://aclanthology.org/2024.acl-long.585/.

Catherine Olsson, Nelson Elhage, Neel Nanda, Nicholas Joseph, Nova DasSarma, Tom Henighan, Ben Mann, Amanda Askell, Yuntao Bai, Anna Chen, et al. In-context learning and induction heads. arXiv preprint arXiv:2209.11895, 2022.

Ori Press, Andreas Hochlehnert, Ameya Prabhu, Vishaal Udandarao, Ofir Press, and Matthias Bethge. Citeme: Can language models accurately cite scientific claims? Advances in Neural Information Processing Systems, 37:7847–7877, 2024.

Rafael Rafailov, Archit Sharma, Eric Mitchell, Christopher D Manning, Stefano Ermon, and Chelsea Finn. Direct preference optimization: Your language model is secretly a reward model. Advances in neural information processing systems, 36:53728–53741, 2023.

Justin T Reese, Leonardo Chimirri, Yasemin Bridges, Daniel Danis, J Harry Caufield, Michael A Gargano, Carlo Kroll, Andrew Schmeder, Fengchen Liu, Kyran Wissink, et al. Systematic benchmarking demonstrates large language models have not reached the diagnostic accuracy of traditional rare-disease decision support tools. medRxiv, pp. 2024–07, 2025.

Weijia Shi, Xiaochuang Han, Mike Lewis, Yulia Tsvetkov, Luke Zettlemoyer, and Wen-tau Yih. Trusting your evidence: Hallucinate less with context-aware decoding. In Kevin Duh, Helena Gomez, and Steven Bethard (eds.), Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 2: Short Papers), pp. 783–791, Mexico City, Mexico, June 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.naacl-short.69. URL https://aclanthology.org/2024.naacl-short.69/.

Shuzheng Si, Haozhe Zhao, Cheng Gao, Yuzhuo Bai, Zhitong Wang, Bofei Gao, Kangyang Luo, Wenhao Li, Yufei Huang, Gang Chen, et al. Teaching large language models to maintain contextual faithfulness via synthetic tasks and reinforcement learning. arXiv preprint arXiv:2505.16483, 2025.

Maojia Song, Shang Hong Sim, Rishabh Bhardwaj, Hai Leong Chieu, Navonil Majumder, and Soujanya Poria. Measuring and enhancing trustworthiness of LLMs in RAG through grounded attributions and learning to refuse. In The Thirteenth International Conference on Learning Representations, 2025. URL https://openreview.net/forum?id=Iyrtb9EJBp.

ZhongXiang Sun, Xiaoxue Zang, Kai Zheng, Jun Xu, Xiao Zhang, Weijie Yu, Yang Song, and Han Li. ReDeEP: Detecting hallucination in retrieval-augmented generation via mechanistic interpretability. In ICLR 2025 Spotlight, October 2024.

Liyan Tang, Philippe Laban, and Greg Durrett. MiniCheck: Efficient fact-checking of LLMs on grounding documents. In Yaser Al-Onaizan, Mohit Bansal, and Yun-Nung Chen (eds.), Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing, pp. 8818–8847, Miami, Florida, USA, November 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.emnlp-main.499. URL https://aclanthology.org/2024.emnlp-main.499/.

AI-Generated

Santosh T.y.s.s, Youssef Tarek Elkhayat, Oana Ichim, Pranav Shetty, Dongsheng Wang, Zhiqiang Ma, Armineh Nourbakhsh, and Xiaomo Liu. CoCoLex: Confidence-guided copy-based decoding for grounded legal text generation. In Wanxiang Che, Joyce Nabende, Ekaterina Shutova, and Mohammad Taher Pilehvar (eds.), Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), pp. 19002–19018, Vienna, Austria, July 2025. Association for Computational Linguistics. ISBN 979-8-89176-251-0. URL https://aclanthology.org/2025.acl-long.931/.

Wenhao Wu, Yizhong Wang, Guangxuan Xiao, Hao Peng, and Yao Fu. Retrieval Head Mechanistically Explains Long-Context Factuality. In The Thirteenth International Conference on Learning Representations, October 2024.

Rongwu Xu, Zehan Qi, Zhijiang Guo, Cunxiang Wang, Hongru Wang, Yue Zhang, and Wei Xu. Knowledge conflicts for LLMs: A survey. In Yaser Al-Onaizan, Mohit Bansal, and Yun-Nung Chen (eds.), Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing, pp. 8541–8565, Miami, Florida, USA, November 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.emnlp-main.486. URL https://aclanthology.org/2024.emnlp-main.486/.

Yuheng Zha, Yichi Yang, Ruichen Li, and Zhiting Hu. AlignScore: Evaluating factual consistency with a unified alignment function. In Anna Rogers, Jordan Boyd-Graber, and Naoaki Okazaki (eds.), Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), pp. 11328–11348, Toronto, Canada, July 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.acl-long.634. URL https://aclanthology.org/2023.acl-long.634/.

Haopeng Zhang, Xiao Liu, and Jiawei Zhang. Extractive summarization via ChatGPT for faithful summary generation. In Houda Bouamor, Juan Pino, and Kalika Bali (eds.), Findings of the Association for Computational Linguistics: EMNLP 2023, pp. 3270–3278, Singapore, December 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.findings-emnlp.214. URL https://aclanthology.org/2023.findings-emnlp.214/.

Huanyu Zhang, Ying Xiao, Xinyue Zhao, Zhuang Tian, Shu-yang Zhang, and Dong Dong. Physicians' knowledge on specific rare diseases and its associated factors: a national cross-sectional study from china. Orphanet Journal of Rare Diseases, 17(1):120, 2022.

Yue Zhang, Yafu Li, Leyang Cui, Deng Cai, Lemao Liu, Tingchen Fu, Xinting Huang, Enbo Zhao, Yu Zhang, Yulong Chen, et al. Siren's song in the ai ocean: A survey on hallucination in large language models. Computational Linguistics, pp. 1–46, 2025.

Penghao Zhao, Hailin Zhang, Qinhan Yu, Zhengren Wang, Yunteng Geng, Fangcheng Fu, Ling Yang, Wentao Zhang, and Bin Cui. Retrieval-augmented generation for ai-generated content: A survey. CoRR, abs/2402.19473, 2024. doi: 10.48550/ARXIV.2402.19473. URL https://doi.org/10.48550/arXiv.2402.19473.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. Judging llm-as-a-judge with mt-bench and chatbot arena. In Proceedings of the 37th International Conference on Neural Information Processing Systems, NIPS '23, Red Hook, NY, USA, 2023. Curran Associates Inc.

Wenxuan Zhou, Sheng Zhang, Hoifung Poon, and Muhao Chen. Context-faithful prompting for large language models. In Houda Bouamor, Juan Pino, and Kalika Bali (eds.), Findings of the Association for Computational Linguistics: EMNLP 2023, pp. 14544–14556, Singapore, December 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.findings-emnlp.968. URL https://aclanthology.org/2023.findings-emnlp.968/.

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